look at Validation certificates first. The simplest one here is Domain Validated (DV) SSL which validates a domain so that the administrator is aware of it and has to actively approve the certificate on the server side either via email or DNS. Then there are Organization Validated (OV) SSL certificates which is only issued after the domain owner proves that an organization (such as Facebook or Google) operates the domain and that their establishment is verified by the CA. This level of validation requires you to provide and verify information such as name, city, state and country of the organization. You can take this one step further with an Extended Validation (EV) certificate which requires more legal paperwork to link the domain with a business. You can easily identify EV certificates by their green bar in the browser featuring the name of the company beside the padlock symbol.

The second type of SSL certificates are those issued based on secured domains. This one is fairly easy to comprehend. There are Single-name SSL certificates which protects a single domain such as www.heythere.com. Then there are Wildcard SSL certificates which protects an unlimited number of subdomains attached to a single domain. So you could have a single certificate for *.heythere.com and it will protect any subdomain attached to heythere.com such as abc.heythere.com and def.heythere.com and so on. And lastly, there are Multi-Domain SSL certificates which can secure a combination of different domains under the same certificate. So you can have heythere.com and whaddap.com secured by the same certificate.

Since there is a significant amount of work needed to issue certain certificates, they end up becoming quite expensive. Even with a lather of marketing mumbo jumbo, CA break up these SSL categories even further into price based tiers with their own nomenclature, thus, adding a lot more complexity to an already complex set up. An EV certificate from GoDaddy can cost you `6,499/year which is a pretty significant investment for someone starting out their own website. And that's with 50% discount, you'll be paying `13,000 upon renewal. So SSL certificates are expensive but they go a long way towards ensuring a better internet. This is where Let's Encrypt comes into the picture. They are a Certificate Authority that provide free certificates and make the entire process of implementing TLS for your domain really simple. It was founded by the Electronic Frontier Foundation, Mozilla Foundation and the University of Michigan with the aim of eliminating payment and reducing the complexity of setting up and maintaining HTTPS. The only catch is that you need to renew the certificate every 90 days as that's how long Let's Encrypt certificates are valid for. However, this is a pretty easy barrier to get across as you can set up a simple cron job to perform renewal every 89 days.

Setting up Let's Encrypt

If you have access to your webserver then you can simply install an TLS certificate using the Certbot ACME(Automated Client Management Environment) client. This way, you can be done with just a few commands. Or if your hosting provider has a built in script to implement Let's Encrypt then you're set. However, since a lot of hosting providers also provide SSL certificates, they don't like the idea of giving up a revenue stream just so that they can serve their customers better. So you're unlikely to find many hosting providers supporting out-of-the-box Let's Encrypt implementation. We ended up using Digital Ocean for setting up HTTPS on a domain as all the other hosting providers we came across had "issues" with setting up Let's Encrypt.

Set up requirements

You can visit the Certbot website for detailed instructions based on which web server software you are using and what OS it is running on. Our droplet on DigitalOcean was running Ubuntu 16.04 with Apache as the web server software. You should have access to the web server via Shell and have all the initial set up kinks ironed out.

Then you'll want to point a Domain to your hosting using the Nameservers that you'll be assigned when you purchased the hosting. If you do this ahead of time then DNS propogation will have taken place by the time you can proceed with installing the certificate. This can take up to 48 hours but we only had to wait two hours.

The first step is to install the Let's Encrypt client i.e. Certbot. Start off by updating the package indexes by typing the following command.

```
$sudo apt-get update
```

And once that's done, you need to install Certbot by typing in.

```
$sudo apt-get install python-letsencrypt-apache
```

If your hosting provider can't find the Certbot client then you'll need to add their repository before using the apt-get install command.

Installing SSL certificate

After the Certbot client has finished the installation process, you can proceed to setting up the SSL certificate. Use the following command to get that done.

```
sudo letsencrypt --apache -d yourdomainname.com
```

It should be pretty self explanatory as to what arguments are being passed here but we'll go ahead and spell it out any ways.